Worksheet 2

Exercise 1: Children's Toys

Imagine a study with children in which the influence of gender (boy, girl) on the color (black, blue, pink) and type of toy (car and dolls) is examined. Girls tend to play with dolls rather than cars and prefer the color pink. Furthermore, it is generally known that the color and type of toy depend on each other: cars are more likely to be black and blue, while dolls are more likely to be dressed in pink.

a) Draw the causal diagram for the three variables `gender`, `color` and `toy`.

b) The data.frame `toys` in the file `toys.rda` contains data from this study. Build a Bayesian network using the causal diagram from a) and the data.

c) Calculate the two conditional probability:
i.e. P(Toy = car | Gender = girl) and P(Toy = car | Gender = boy).
Describe in your own words how these probabilities should be interpreted.

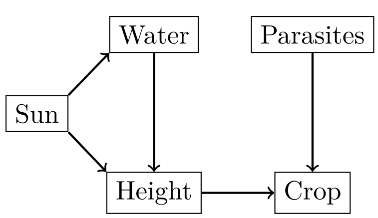
d) Calculate the two joint probabilities
P(Toy = car, Gender = girl) and P(Toy = car, Gender = boy).
What is the difference compared to the conditional probabilities in b)?

e) Describe the meaning of the probability P(Toy = car | do(Gender = girl))? Why is it impossible to observe data for this?

f) Which paths transport causal effect from gender on toy? Justify your answer.

Exercise 2: Apple tree

Have a look at the example from the lecture on the crop of apple trees.



We assume the following discrete values for the five random variables:

- Sun $\in$ {little, much}
- Water $\in$ {little, much}
- Height $\in$ {little, much}
- Crop $\in$ {little, much}
- Parasites $\in$ {yes, no}

The following probabilities are given:

- P(Sun = much) = 0.7
- P(Water = much | Sun = much) = 0.05 and
  P(Water = much | Sun = little) = 0.65.
- P(Height = much | Sun = much, Water = much) = 0.4,
  P(Height = much | Sun = little, Water = much) = 0.45,
  P(Height = much | Sun = much, Water = little) = 0.55,
  P(Height = much | Sun = little, Water = little) = 0.25.
- P(Parasites = yes) = 0.18
- P(Crop = high | Height = much, Parasites = yes) = 0.67,
  P(Crop = high | Height = little, Parasites = yes) = 0.01,
  P(Crop = high | Height = much, Parasites = no) = 0.9,
  P(Crop = high | Height = little, Parasites = no) = 0.42.

a) Build a Bayesian network with R using the given conditional probabilities. Visualize the resulting causal diagram.

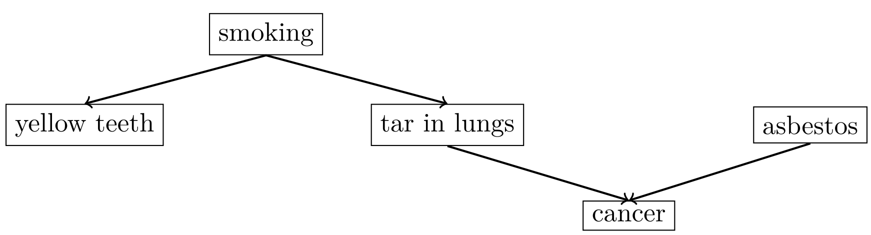
b) Calculate the marginal probabilities of a high crop yield, P(Crop = high), and a low crop yield, P(Crop = low), using R.

c) What is the probability of a high crop yield given that a tree has parasites?

d) Are Height and Parasites independent given the Crop?

Exercise 3: Lung cancer

Have a look at the following causal graphical model about lung cancer.



Which of the following independence statements hold?

1. `yellow teeth` is independent of `tar in lungs` conditioning on `smoking`.
2. `tar in lungs` is independent of `asbestos`.
3. `tar in lungs` is independent of `asbestos` conditioning on `cancer`.
4. `smoking` is independent of `cancer` conditioning on `tar in lungs`.
5. `yellow teeth` is independent of `tar in lungs`.